PROJECT DESIGN PHASE-I

Solution Architecture

Date	18 February 2026
Team ID	
Project Name	Malaria Detection using Deep Learning
Maximum marks	4

Solution Architecture

The solution architecture defines the structural design of the AI-Based Malaria Detection System. It bridges the identified healthcare problem with a scalable and efficient technical implementation. The architecture outlines system components, data flow, and deployment strategy.

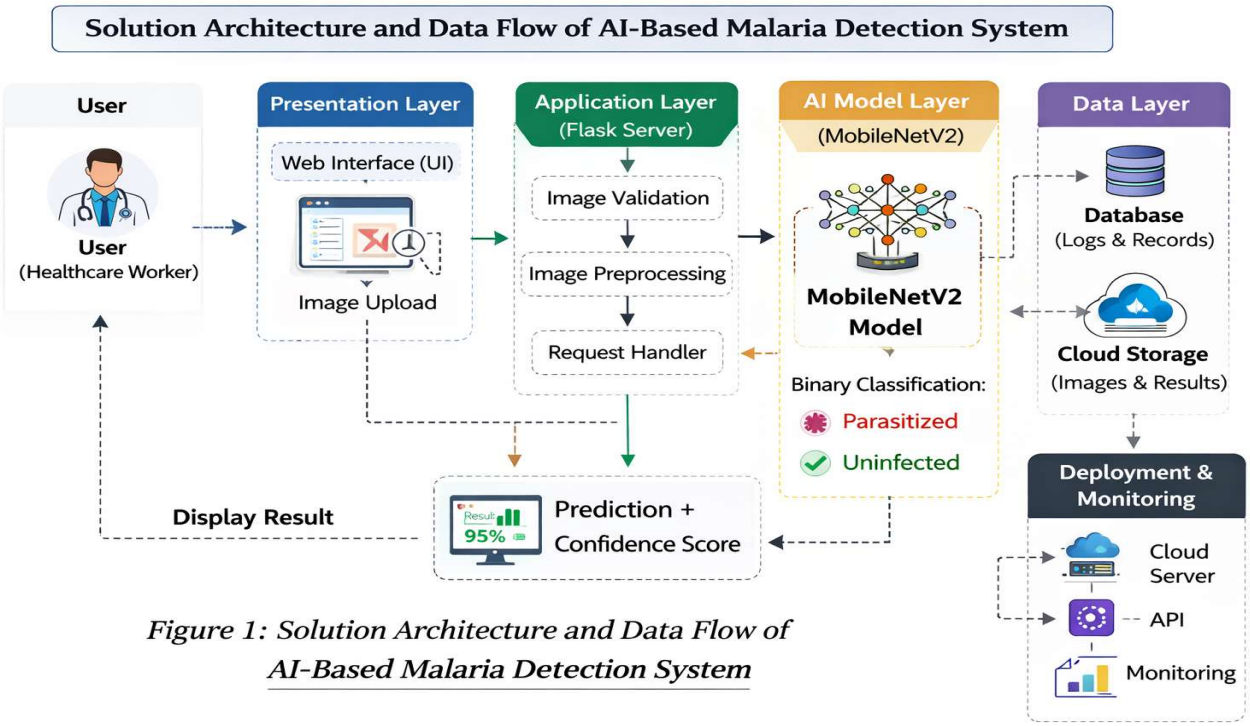

Figure 1: Solution Architecture and Data Flow of AI-Based Malaria Detection System

Component Description

1. Presentation Layer

The Presentation Layer provides the user interface for healthcare workers to interact with the system. It allows users to upload blood smear images and view diagnostic results along with confidence scores. The interface is web-based, ensuring accessibility across devices without requiring additional software installation.

2. Application Layer (Flask Server)

The Application Layer handles the business logic of the system. It receives image uploads, validates file formats, performs preprocessing (resizing, normalization), and forwards the processed image to the AI model. It also manages request handling and sends prediction results back to the user interface.

3. AI Model Layer (MobileNetV2)

The AI Model Layer contains the pre-trained MobileNetV2 deep learning model using transfer learning. It performs binary classification of the uploaded blood smear image into:

- Parasitized
- Uninfected

The model outputs a probability score indicating prediction confidence. MobileNetV2 is selected for its lightweight architecture and efficient performance in low-resource environments.

4. Data Layer

The Data Layer manages storage and logging functions. It can store:

- Uploaded images (optional)
- Prediction results
- Confidence scores
- System logs

This layer enables auditing, performance tracking, and future model retraining.

5. Deployment & Monitoring Layer

The Deployment Layer ensures system availability and scalability. The solution can be deployed on a cloud server or local hospital infrastructure. Monitoring tools track system performance, response time, and usage patterns to ensure reliability and continuous improvement.